

Cancer	Rank	Peak (cm ⁻¹)	Representative assignment	Candidate assignments	Feature importance
LUN	1	539	S-S stretching	S-S stretching; disulfide-rich protein; cysteine/cystine-related mode	5.76
LUN	2	1000	Phenylalanine	Phenylalanine; urea; aromatic ring breathing	5.70
LUN	3	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	5.61
LUN	4	849	Tyrosine	Tyrosine; protein ring mode; collagen/proline-related mode	4.98
CRC	1	795	Hippuric acid	Hippuric acid; protein C-C/C-N mode; ring breathing mode	10.50
CRC	2	618	C-S stretching	C-S stretching; protein sulfur-containing residues; creatinine-related mode	6.24
CRC	3	1000	Phenylalanine	Phenylalanine; urea; aromatic ring breathing	6.14
CRC	4	448	Ring/skeletal deformation	Ring/skeletal deformation; protein or nucleic-acid ring modes	5.46
BLC	1	1000	Phenylalanine	Phenylalanine; urea; aromatic ring breathing	13.64
BLC	2	683	Creatinine	Creatinine; nucleic-acid ring mode; amino-acid side-chain mode	8.41
BLC	3	895	Uric acid	Uric acid; nucleic-acid backbone/ring mode; C-C skeletal mode	6.24
BLC	4	849	Tyrosine	Tyrosine; protein ring mode; collagen/proline-related mode	5.38
PRO	1	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	7.61
PRO	2	1651	Amide I / protein	Amide I / protein; C=C lipid stretching; water/protein carbonyl mode	7.59
PRO	3	934	Creatinine	Creatinine; protein C-C skeletal mode; carbohydrate-related mode	7.03
PRO	4	1449	CH2 deformation / lipid	CH2 deformation / lipid; protein CH2/CH3 bending; collagen-related mode	6.51
OVA	1	1148	Uric acid C-N stretching	Uric acid C-N stretching; protein C-N mode; nucleic-acid ring mode	10.29
OVA	2	795	Hippuric acid	Hippuric acid; protein C-C/C-N mode; ring breathing mode	8.32
OVA	3	934	Creatinine	Creatinine; protein C-C skeletal mode; carbohydrate-related mode	6.51
OVA	4	1651	Amide I / protein	Amide I / protein; C=C lipid stretching; water/protein carbonyl mode	5.71
PAN	1	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	5.79
PAN	2	1651	Amide I / protein	Amide I / protein; C=C lipid stretching; water/protein carbonyl mode	5.65
PAN	3	1148	Uric acid C-N stretching	Uric acid C-N stretching; protein C-N mode; nucleic-acid ring mode	5.50
PAN	4	448	Ring/skeletal deformation	Ring/skeletal deformation; protein or nucleic-acid ring modes	4.74